Smart AI Solution for Rostelecom Advertisement Detection

Binary Classification of Rostelecom ads from images

Muhammad Mujahid Haruna – AIJC 2025 Main Stage

Problem Overview

- Automatically detect **Rostelecom ads** in submitted photos.
- Reduce manual checking effort while maintaining high accuracy.

• **Challenge:**
- Imbalanced dataset (fewer positive ads).
- Variations in placement, background, lighting, and angle.

• **Requirement / Goal:**
- Achieve **high F1-score** for leaderboard ranking.

Insights from Data

- Class 1 (Rostelecom ads) mostly appear in elevators or entrance walls.

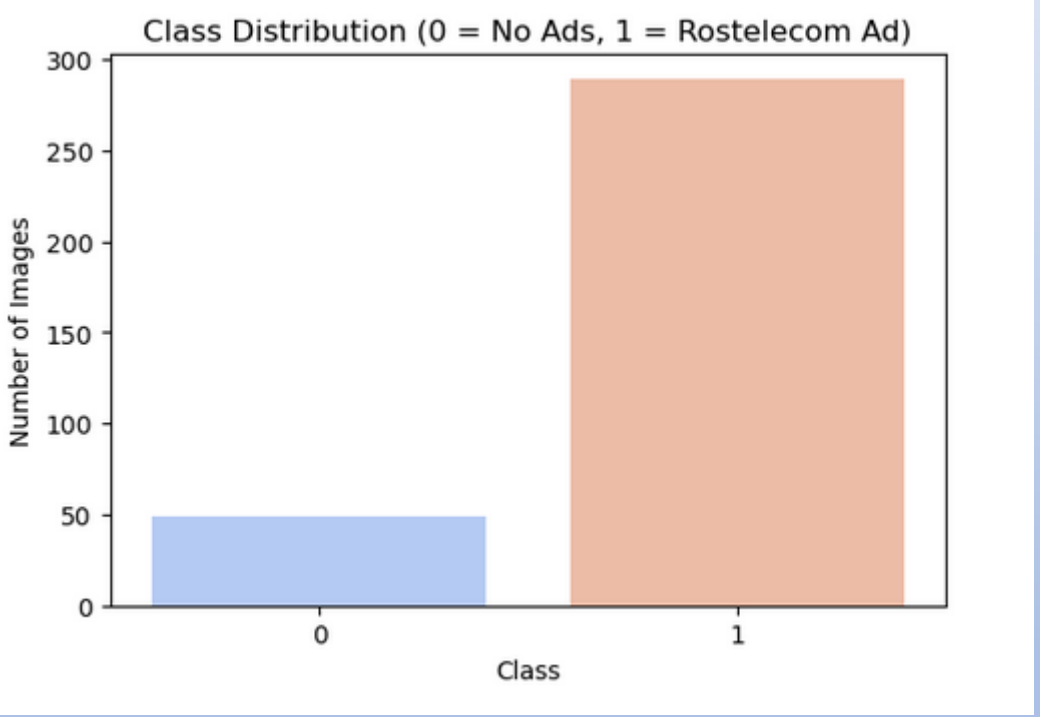
- Dataset is imbalanced:

  💡 Solution: Use class weights during training to ensure fairness & optimize F1-score.

- Variation in size, angle, and brightness for more generalization.

  💡 Solution: Data augmentation helps the model generalize better.

- Model usually find it difficult to classify images where the Rostelecom ads is far.



Preprocessing

- Image normalization and resizing (224x224) for model stability.
- Data augmentation to generalize over variations:
  - Slight rotation, shifts, shear, zoom.
  - Brightness adjustments limited to realistic range.

💡 Goal: Reduce overfitting, improve robustness.

Class weights applied during training to balance the positive/negative classes.

💡 Outcome: Optimized F1-score and fair evaluation.

Model Development & Approach

- Tested multiple small CNN architectures to identify optimal performance.

- Observed common issues: underfitting, overfitting, unstable validation accuracy.

💡 Approach: Use Keras Tuner for hyperparameter optimization to systematically select the best CNN.

📌 Best CNN selected after hyperparameter tuning based on validation performance.

📌 Final model achieves high accuracy while avoiding overfitting.

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 220, 220, 32)	2,432
conv2d_1 (Conv2D)	(None, 216, 216, 16)	12,816
flatten (Flatten)	(None, 746496)	0
dense (Dense)	(None, 16)	11,943,952
dense_1 (Dense)	(None, 1)	17

Submission & Results

- Generated submission.csv containing predicted classes for test images.
- Format: image filename + predicted class (0/1)
- 📌 Ready for leaderboard evaluation

📌 Code is clean, well-commented, and reproducible in Google Colab
📌 Smart data handling: class weights & augmentation applied
📌 Validation scheme ensures fair evaluation💡 Outcome: Solution optimized for maximum F1-score & top leaderboard ranking.

```
🎯 Making predictions on test set...
29/29 ━━━━━━━━━━━━━━━━━━━━ 41s 1s/step

✅ Submission file saved as 'submission_final.csv'
Ready for upload to leaderboard!
```